

Enter The Matrix!

Everyone and their uncles have hopped on to the LCD bandwagon, but how many of us know about the different kinds of LCDs out there?

Michael Browne

No two LCDs are created alike

That may sound dubious at first, but as you'll see, it's very true. The differences between LCD technologies are fundamental, based on the actual panel matrix that the LCD (Liquid Crystal Display) uses.

Each type of panel is designed to suit a particular usage pattern—gaming, movies, image editing, etc. This is actually intentional. As far as history is concerned, compared to CRTs, which have been around since 1897, LCDs are rather new products, and constant innovations are being made.

If you're going out to buy an LCD, just reading manufacturer specifications isn't going to clear the fog. If you want to know whether a particular LCD is suited for your needs, you're going to have to understand more about the manufacturing technologies. Almost all LCDs made today are the jack-of-all-trades kind, and the only way to discern a panel's suitability to a task is to check the type of matrix used.

There are broadly four types of LCD matrices being manufactured today:

1. TN (Twisted Nematic) also called TN+Film
2. IPS (In Plane Switching)
3. MVA (Multi-domain Vertical Alignment)
4. PVA (Patterned Vertical Alignment)

TN / TN+Film

TN panels are the adults of the LCD world. They've been on the scene since the time of Passive Matrices. The term "Twisted Nematic"

actually refers to the way the liquid crystals are organised.

How A TN Panel Works

When voltage is applied (see Figure 1) you'll notice the outer set of crystals hardly change position (mostly parallel to the panel's surface). All the while, the inner crystals orient themselves in such a way that their axis is perpendicular to the panel's surface—they twist. The problem in reality is the outer crystals aren't really parallel to the surface but *nearly* parallel. If viewed as a whole (the entire matrix), there are noticeable irregularities. These irregularities cause TN panels to have a characteristically poor Contrast Ratio, which is immediately noticeable while viewing multimedia content. Also, the viewing angles are adversely affected because of such irregularities.

Another disadvantage that TN panel-based LCDs face is the fact that they're natively capable of displaying only 262,144 colours—18-bit colour as opposed to 24-bit colour. TNs are basically 6-bit panels—the 18 bits comprise of 6 bits per colour component, red, green, and blue. However, manufacturers get to the magical figure of 16.7 million colours (24-bit) by a technique called *dithering*. Dithering is simply a technique for altering the values of adjacent dots (or pixels for that matter) on a matrix so as to create the illusion of a larger gamut of colours than what actually exists.

The advantages that TN panels have is that they're by far the fastest of all the technologies in terms of response times. This, coupled with the fact that TN panel LCDs are the cheapest to manufacture, is the reason that when you go out to buy an LCD, the majority of manufacturers will supply you with a model containing a panel based on this technology.

